Auditing Process Knowledge for Parsimonious and Explainable Industrial Cyberattack Detection

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Project Website

GitHub Repository

Motivation and Contribution

Research question: Which engineering relationships should be admitted into a hybrid cyberattack detector?

- Industrial control systems (ICSs) need temporal attack detection that operators can interpret. We audit engineering relationships before adding them to a detector.
- Six relationships pass every audit check and form the conservative CORE. We combine their scores with a 25-variable LSTM.
- **Key finding: more process knowledge is not better. CORE (6 relationships) beats using all 16 retained.**

Process-Knowledge Audit: Four Screening Criteria

What is a process-knowledge audit? Engineers can write many physically plausible rules about a plant (e.g., "if valve MV201 opens, flow FIT201 should rise"), but some are rarely observed, unstable over time, or redundant. The audit tests every candidate rule against normal operating data and admits only reliable ones into the detector as deviation scores.

1 Empirical support. Does the relationship show a clear, measurable effect in TRAIN data (effect size, correlation, enough samples)?

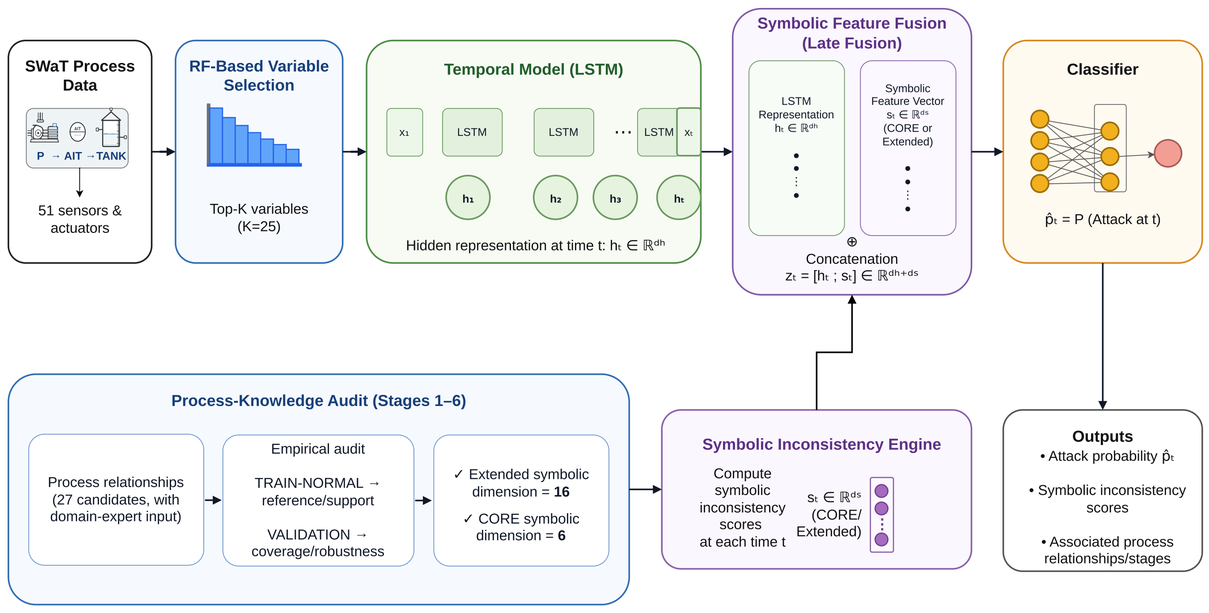
2 Validation coverage. Does it apply to at least 90% of VALIDATION samples and score low on normal data? If not, it is labeled Limited.

3 Robustness. Is it stable from TRAIN to VALIDATION, with well-represented normal data and a time lag inside the tested range?

4 Redundancy. Does it add information that other retained relationships do not already provide?

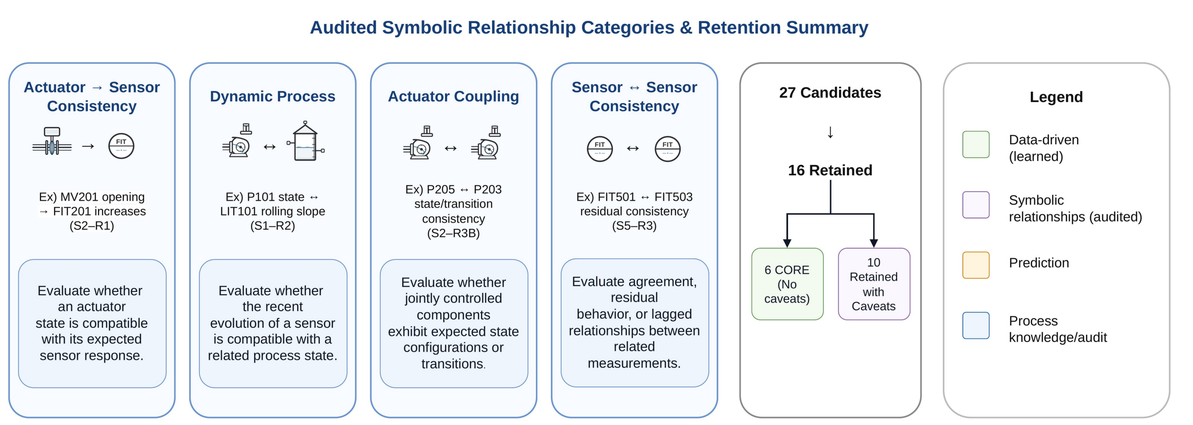
27 candidates → 6 CORE + 10 retained with caveats + 2 limited + 9 rejected

Architecture and Hybrid Detection Framework



LSTM: 120 observations (approximately 120 s), 64 hidden units. Final classifier: linear layer + sigmoid.

Audited Relationship Categories and Retention



How to read: each candidate relationship falls into one of four categories and is turned into a score of how far the plant departs from it. **Six pass every check (CORE). Ten pass with documented caveats.** Together, the 16 retained form the Extended set.

Method Details

Data and Chronological Protocol

- **SWaT:** 51 sensors and actuators across six water-treatment stages; 449,919 attack-collection observations with 35 attack events.
- **Chronological split:** boundaries fall in normal operation, so no attack event spans two partitions.
- **TEST stays held out:** scaling, variable ranking, audit, tuning, and threshold use TRAIN and VALIDATION only.

Temporal Branch (LSTM)

- **Variable selection:** Random Forest (500 trees) ranks the 51 variables on TRAIN; K = 25 is chosen on VALIDATION (94.6% of importance).
- **Model:** windows of L = 120, stride 1; one LSTM layer, 64 hidden units, dropout 0.2.
- **Training:** Adam (10^{-3}), batch 256, up to 30 epochs, early stopping, weighted loss; five random seeds.

Symbolic Branch and Fusion

- **Scores:** each retained relationship gives a deviation score; larger means further from normal behavior.
- **Fusion:** the window's final scores (6 for CORE, 16 for Extended) are concatenated with the 64-d LSTM output, then a linear layer and sigmoid give the attack probability.
- **Threshold:** chosen on VALIDATION to maximize MCC, then fixed for TEST.

Held-Out Detection Results

Fixed chronological TEST partition. Mean ± standard deviation over five seeds.

Model	Raw / symbolic	MCC	F1	ROC-AUC
Full LSTM	51 / 0	0.135 ± 0.082	0.066 ± 0.043	0.621 ± 0.084
Selected LSTM	25 / 0	0.183 ± 0.101	0.137 ± 0.111	0.520 ± 0.049
NS CORE	**25 / 6**	**0.247 ± 0.063**	**0.161 ± 0.063**	**0.600 ± 0.052**
NS Extended	25 / 16	0.155 ± 0.123	0.084 ± 0.098	0.577 ± 0.041

CORE has the highest mean TEST MCC and F1 among these four models. Full LSTM has the highest mean ROC-AUC.

CORE vs Selected LSTM: paired MCC, F1, and AUC differences are not statistically significant across five seeds ($p \geq 0.0625$).

Chronological Evaluation

TRAIN: 270,000 observations, 22 attacks
VALIDATION: 68,319 observations, 3 attacks
TEST: 111,600 observations, 10 attacks

No sequence or attack crosses a split boundary. TEST remains held out during development.

Generalization over time: mean MCC

Operator-Facing Process Evidence

Six CORE relationships identify where process behavior deviates:

Stage	Relationship	Evidence type
S1	MV101–FIT101	Valve–flow consistency
S1	P101–LIT101	Pump–level dynamics
S1	FIT101–LIT101	Flow–level dynamics
S2	MV201–FIT201	Valve–flow consistency
S2	P205–P203	Actuator coupling
S5	FIT501–FIT503	Sensor residual consistency

Limitations and Next Steps

- All four models miss six of ten TEST attack intervals, covering 78.4% of attack observations. Absolute performance remains weak and is not deployment-ready.
- A symbolic deviation indicates abnormal process behavior. It cannot distinguish an attack from equipment or sensor failure.
- Next: evaluate other processes and operating modes, accounting for attack severity and detection delay relative to physical timescales.

Parsimony reduces the number of engineering relationships. Chronological generalization remains unresolved.